Project 1: Counterfactuals & Alternative Technical Realities

The Motion Economy Standard — What If Dvorak Had Replaced QWERTY?

I. Historical Pivot & Rationale

Baseline reality (1932–1956): August Dvorak, a University of Washington professor, and William Dealey filed a patent for the Dvorak Simplified Keyboard in 1932, granted in 1936. It placed roughly 70% of typing on the home row (versus about 32% on QWERTY) and claimed to cut a typist's daily finger travel from 12–20 miles down to about one. During WWII, the Navy ran retraining experiments — with Dvorak himself involved in overseeing them — that reported typists recouping their retraining costs in speed gains within just ten days. Despite this, the Navy declined to switch, and a subsequent adoption bid was quietly canceled by the Treasury Department for reasons that were never made clear. Two decades later, a properly controlled 1956 GSA study by Earle Strong found Dvorak typists could only match, not beat, their old QWERTY speeds — killing the layout's chances for good.

The pivot point: 1945. In this counterfactual, the Treasury Department does not cancel the wartime adoption bid. Riding the same wartime urgency that produced the original Navy results, the federal government formally adopts Dvorak as the mandatory keyboard standard for all government typists starting in 1945 — and, because the government is by far the largest single purchaser of office equipment and the largest trainer of clerical labor, private industry and education follow within a decade.

Why this pivot works well: it's a documented near-miss with a real, unexplained cancellation at its center — a genuine gap in the record for the counterfactual to fill. It's also the textbook case economists use for path dependence and technological lock-in (Paul David's 1985 paper "Clio and the Economics of QWERTY"), so there's a real theoretical framework to cite. And because Dvorak's whole design philosophy was about minimizing wasted physical motion, the counterfactual naturally extends past keyboards into a much bigger question: what happens if "motion economy" becomes a governing design principle for an entire generation of American engineering, right as the Cold War space race is getting underway?

II. STEEP-V Analysis

Vector	Implication in the Dvorak timeline
Society	"Motion economy" becomes a mainstream value taught from grade school on, alongside penmanship. Efficient movement is treated as a personal virtue, not just a workplace metric.
Technology	Dvorak's home-row philosophy generalizes past keyboards into a broader discipline of "motion-efficient design" — applied to car dashboards, telephone keypads, and eventually aerospace cockpit layouts a generation before real-world human factors engineering matured.
Economic	"Human Factors Engineer" becomes a licensed, high-status profession by the late 1950s. A federal certification industry grows up around it, the way safety and emissions certification exist today.
Environmental	Marginal — lower repetitive-strain injury rates reduce workers'-compensation costs, a quiet but real economic ripple.
Political	The wartime Navy program formalizes into a permanent federal agency — a **Bureau of Motion Economy** — that certifies government-procured equipment (from typewriters to weapons-control panels) against a mandated "motion economy index."
Values	Efficiency of motion becomes tied to Cold War-era national productivity and competitiveness — "wasted motion is wasted patriotism."

III. Phase 1 (15 min) — Mapping the Signal Across Dator's 4 Arcs

Weak signal: the 1945 federal Dvorak mandate, read forward as "motion economy" becoming a governing design doctrine.

Arc	Trajectory
Growth	Dvorak becomes the universal keyboard standard by the 1960s. Typing-speed records shatter; "keyboard efficiency" is celebrated as a national productivity metric and taught from childhood.
Collapse	Retraining millions of QWERTY-trained clerks and secretaries triggers a "Great Retraining Crisis" — office productivity craters for years, backlogs pile up, and some agencies quietly keep illegal QWERTY machines running as a shadow system.
Discipline	The government mandates strict, tiered "keyboard literacy" certification tied to civil-service rank; compulsory retraining bootcamps and inspection regimes police the transition nationwide.
Transformation	Dvorak's motion-economy philosophy sparks an early, government-driven human factors design revolution — reshaping vehicle and equipment design decades before ergonomics became a mainstream field in the real timeline.

Chosen arc for the Time Slice: Transformation — it's the arc that pays off the counterfactual's real premise (Dvorak's core idea was never really about keyboards, it was about minimizing wasted human motion), and it opens onto Cold War-era aerospace design, which is rich, well-documented, and visually distinct territory for a physical artifact.

IV. Phase 2 (20 min) — The 1962 Time Slice

Anchor slice: 1962 — chosen because it lands squarely in the early Mercury/Gemini era, when cockpit ergonomics was already a genuine, high-stakes engineering concern in our own timeline. Placing a federal "motion economy" bureaucracy inside that real historical pressure point makes the speculative institution feel load-bearing rather than decorative.

1. Horizon Scanning (STEEP-V)

See table above — condensed for this slice: motion economy is now a licensed profession, a federal certifying bureaucracy, and a Cold War patriotic value simultaneously.

2. The Infrastructure

Core technology: Dvorak-derived keyboards are the unquestioned standard; motion-economy scoring rigs (grease-pencil motion-path tracing, later stopwatch-and-camera studies) are used to certify new equipment designs.
Regulatory body: The **Bureau of Motion Economy**, spun out of the original wartime Navy program, certifies all federally procured equipment against a mandated "motion economy index" (MEI).
Legacy systems: A shrinking population of pre-1945 QWERTY typewriters survive as collector's heirlooms — illegal for use in any federal office.

3. The Day in the Life

Persona: Walter Kowalski, 34, a design engineer at an aerospace contractor working on Mercury capsule cockpit layout. **Morning routine:** Reviews his team's Motion Economy Index scorecard from last week's cockpit mockup review before heading into a certification hearing.
Primary conflict: His proposed astronaut control-panel layout works perfectly in functional testing but fails Bureau certification on a technicality — a switch cluster sits half an inch outside the mandated motion-economy envelope. The design is rejected days before a critical schedule milestone, forcing an emergency redesign under launch-window pressure.

4. Ripple Effects (2nd & 3rd order)

Primary innovation: Dvorak's "logical motion" philosophy, elevated to federal design doctrine after the 1945 keyboard mandate.
1st order (direct effect): The Bureau of Motion Economy is established to certify federal equipment for motion efficiency.
2nd order (unintended reaction): Certification bureaucracy slows innovation cycles across unrelated fields — aerospace, weapons systems, household appliances — as designers must satisfy motion-economy metrics before functional performance is even tested.
3rd order (cultural backlash): A "Functionalist Revolt" grows among engineers who argue the certification regime rewards measurable motion metrics over real-world performance, leading to unofficial "shadow labs" that build and flight-test uncertified prototypes off the books.

5. Anomalies & Edge Cases

New-normal dark market: Underground shadow labs where engineers test uncertified, "wasteful but effective" designs outside Bureau oversight.
Who's left behind: Engineers whose intuitive, high-performing designs fail rigid motion-economy scoring — and older workers trained before 1945 who never fully requalified and quietly lost seniority.

6. The Extrapolation Leap

Artifact from the future: A **Motion Economy Non-Conformance Notice** — the rejection slip issued to Walter's team, plus a small metal "MEI Certified" equipment plaque as a companion object (see Phase 3).

Headline of the day: *"Bureau of Motion Economy Rejects Mercury Cockpit Mockup, Cites Non-Compliant Switch Cluster — Engineers Call for Reform"*

V. Phase 3 (25 min) — Physical Artifact Prototype

Primary artifact: *The Motion Economy Non-Conformance Notice* — an official rejection form issued by a Bureau inspector at a design review, attached to a small physical "MEI Certified" plaque as the companion object it's meant to accompany (or, in this case, deny).

Materials by zone (cold-fastener, non-art bricolage — no hot glue or clay):

Zone A: Synthetic & Schematic Substrates

LLM interface for generating period-accurate Bureau terminology, MEI scoring language, and 1962 aerospace engineering jargon
Grid-lined index cards and graphite pencil for blocking out the Non-Conformance form and the cockpit switch-cluster diagram it references before committing to final stock

Zone B: Diegetic Artifacts & Hardware Scavenging

Carbon-copy triplicate paper stock (inspector's copy, engineer's copy, file copy) for the Notice itself
A rubber "REJECTED — MEI NON-CONFORMANT" stamp and ink pad
A small aluminum-foil-wrapped cardstock tag as the companion "MEI Certified" equipment plaque, punched and wired on
Disassembled toggle switches or exposed logic-board fragments (e-waste scavenging) glued-free — mounted with binder clips — to mock up the offending "switch cluster" the Notice references
Cold fasteners only: binder clips, rubber bands, masking tape, and transparent takeout-container plastic as a protective sleeve over the plaque

Zone C: Wearable Interfaces & Spatial Scenography

A folded cardstock "Bureau of Motion Economy — Field Inspector" placard or badge, for staging the moment of rejection
Flat-pack cardboard and a desk lamp to stage a 1962 drafting-table scene when photographing the artifact — hard directional light, no ambient wash
Optional: a hand-drawn grease-pencil motion-path diagram (arcs tracing hand movement across the switch cluster) taped beside the rejected mockup, echoing the Bureau's actual scoring method

Layout (Non-Conformance Notice):

```
═══════════════════════════════════════
 BUREAU OF MOTION ECONOMY
 FORM ME-12: NOTICE OF NON-CONFORMANCE
═══════════════════════════════════════
ENGINEER: W. KOWALSKI      DIV: AERO-4
PROJECT: MERCURY CTRL PANEL MK-III
DATE: 06.14.1962   INSPECTOR: R. FENN

FINDING: MOTION ECONOMY INDEX FAILURE
MANDATED MEI:     0.82 (MIN)
MEASURED MEI:     0.71
CAUSE:  SWITCH CLUSTER 4 EXCEEDS
        REACH ENVELOPE BY 0.5 IN

REQUIRED ACTION:
>> RESUBMIT REVISED LAYOUT WITHIN 5 DAYS
>> ATTACH UPDATED MOTION-PATH DIAGRAM
>> NO FLIGHT-HARDWARE FAB UNTIL CLEARED

INSPECTOR SIGNATURE: ___________
═══════════════════════════════════════
```

VI. Phase 4 (15 min) — Contextual Notation

MEI plaque tag (companion object, withheld pending resubmission): a small foil-wrapped tag reading "CERTIFIED — MEI 0.82+ — BUREAU OF MOTION ECONOMY," designed to look like a UL safety sticker, but for ergonomic compliance rather than electrical safety.

Warning disclaimer (printed at the bottom of the Notice pad):

This notice constitutes an official finding under the Federal Motion Economy Standards Act of 1947. No federally funded hardware may proceed to fabrication while a Notice of Non-Conformance is outstanding. Falsifying a Motion Economy Index submission is a violation of Bureau Regulation 6(a).

Operating instructions (reverse side of the form):

Form ME-12 must be countersigned by the project's supervising engineer within 48 hours of issuance. Contractors may appeal a finding to the Bureau's Design Review Board within ten business days. Three non-conformances on a single project within one fiscal year trigger a full program audit.

VII. Theoretical Alignment

A **Transformation-arc** counterfactual: rather than a keyboard layout simply winning a market fight, an entire design philosophy — Dvorak's minimization of wasted motion — becomes federal doctrine and reshapes engineering culture a generation early, which is the more interesting and higher-stakes version of "what if the more efficient technology had won."
Directly engages the real economic literature on **path dependence and lock-in** (David's QWERTY paper), letting the proposal cite an actual scholarly framework rather than inventing one from scratch.
The **Non-Conformance Notice** functions the way the deck's "1:1 scale speculative artifact" examples do (see: *The Eliza Mandate*) — a mundane bureaucratic rejection slip that makes an abstract structural choice (which keyboard won in 1945) suddenly feel personal, procedural, and load-bearing for a real engineer under a real deadline.
Surfaces a structural-bias question consistent with Theme 1's concept map: a certification regime built around a single quantifiable metric (MEI) systematically punishes intuitive, high-performing designs that don't fit the metric — the same failure mode as any optimization target that isn't the thing you actually care about.